

ElevanceSkills Technology Private Limited

No.2/180, Barur, Dharmapuri, Tamil Nadu 635201, India

Project Report

Project Title: Real Time Twitter Analytics Dashboard

Starting Date: 12-08-2026

Ending Date: 13-09-2026

Duration: 1 Month

Domain: Data Analytics

Organization: ElevanceSkills Technology Private Limited

Mentor: Muskan & Satya Prakash Mohanty

Submission Date: 12-09-2026

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Introduction

This project was undertaken as part of a data analytics internship to develop practical skills in **Power BI, data visualization, data filtering, aggregation, and DAX**. The project focused on analyzing a Twitter dataset containing tweet-level information such as impressions, likes, retweets, replies, engagement rate, media interactions, and other engagement-related metrics.

The main purpose of the project was to transform raw Twitter data into meaningful visualizations and insights. Six analytical tasks were completed, covering tweet interactions, engagement rate comparison, media interactions, overall engagement metrics, monthly engagement trends, and top-performing tweets.

Learning Objectives

The major learning objectives of this project were:

- To understand the process of importing and analyzing datasets in Power BI.
- To learn how to clean and prepare data for analysis.
- To create meaningful visualizations from raw data.
- To use **SUM, AVERAGE, COUNT**, and other aggregations.
- To develop and apply **DAX measures**.
- To implement conditional filtering in Power BI.
- To understand date-based and time-based analysis.
- To analyze social media engagement metrics.
- To create interactive and easy-to-understand dashboards.
- To develop practical problem-solving skills while working with real-world data.

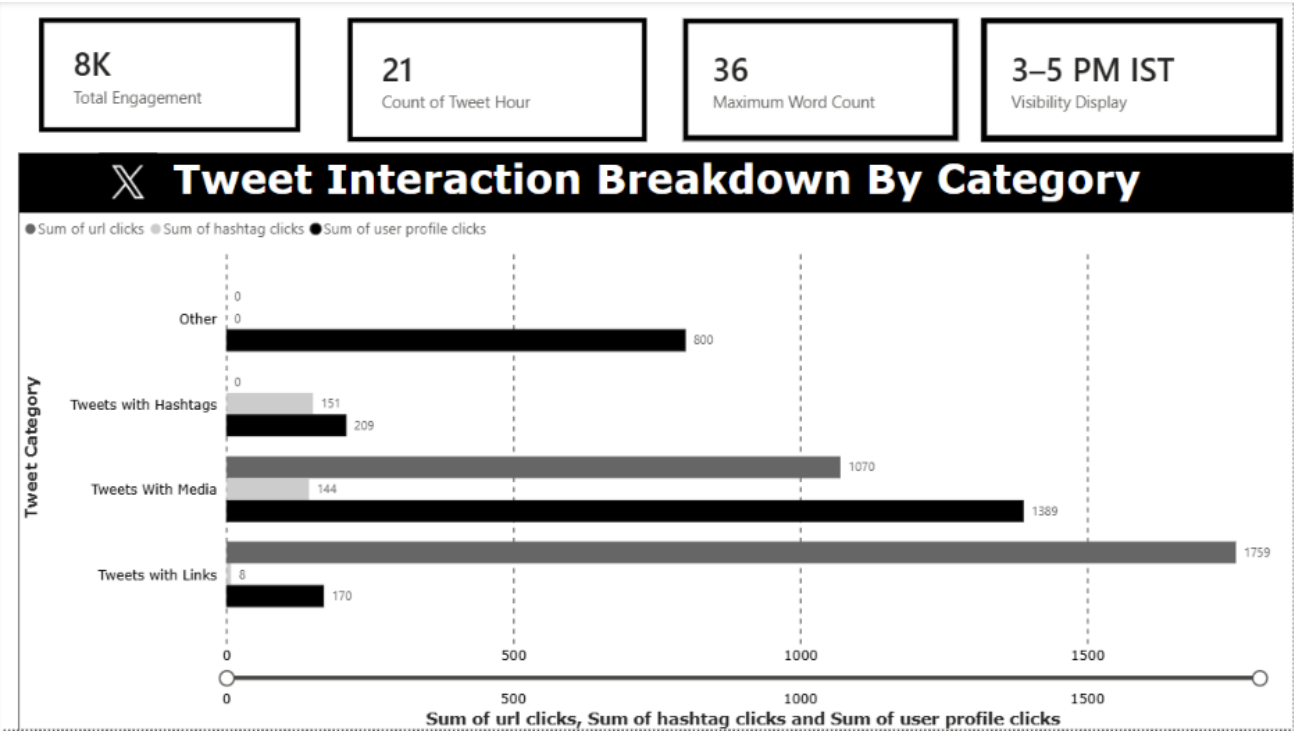
Activities and Tasks

Six major tasks were completed during the project.

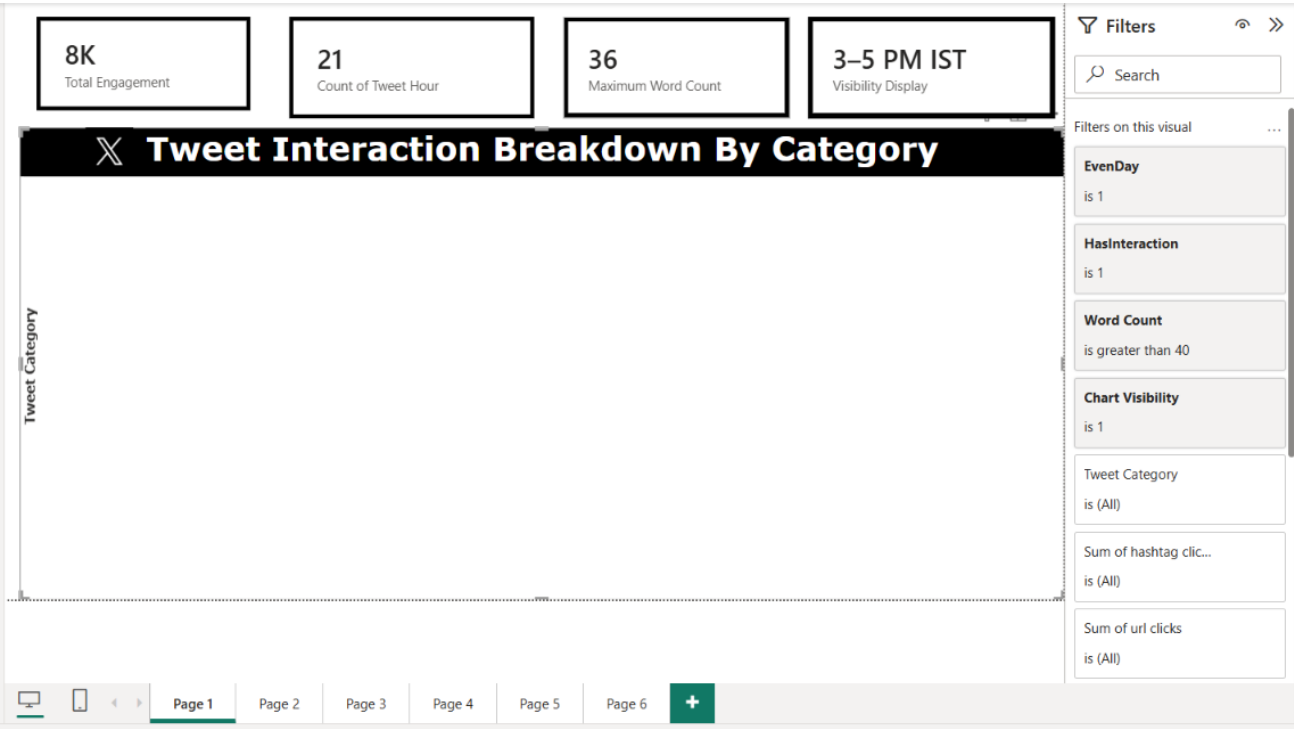
Task 1 – Tweet Interaction Breakdown by Category

A clustered bar chart was created to analyze interactions across different tweet categories.

The analysis included: URL clicks , User profile clicks , Hashtag clicks .



Task (1) Without Filter



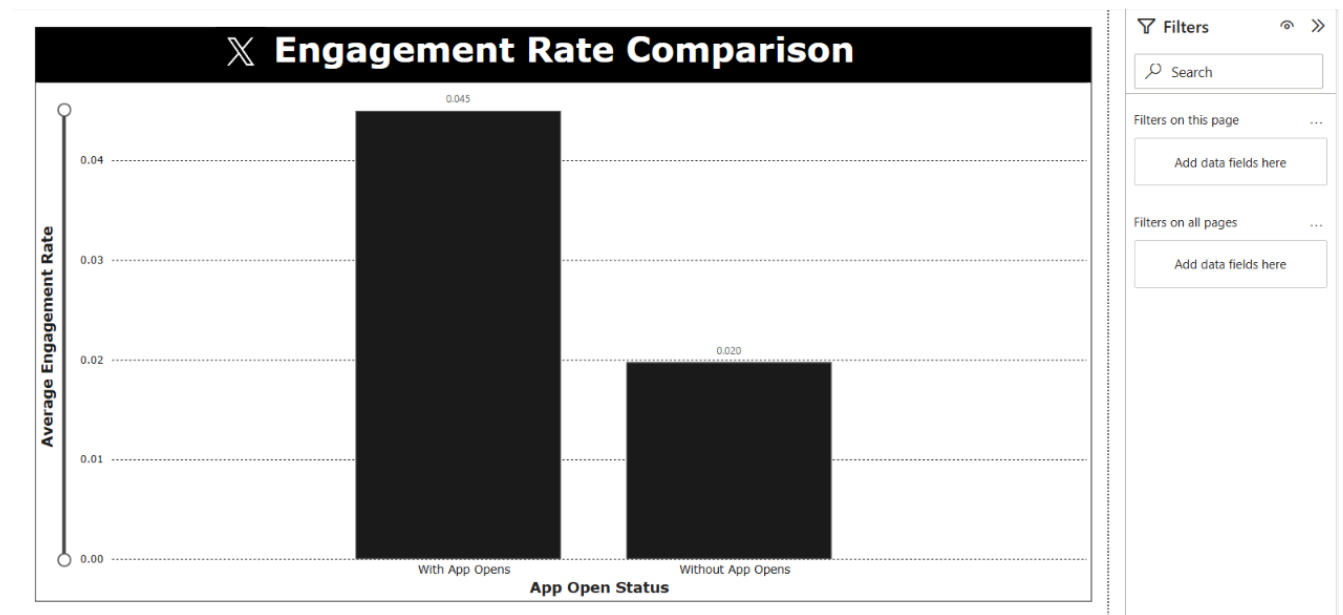
Task (1) With Filter

Task 2 – Engagement Rate Comparison

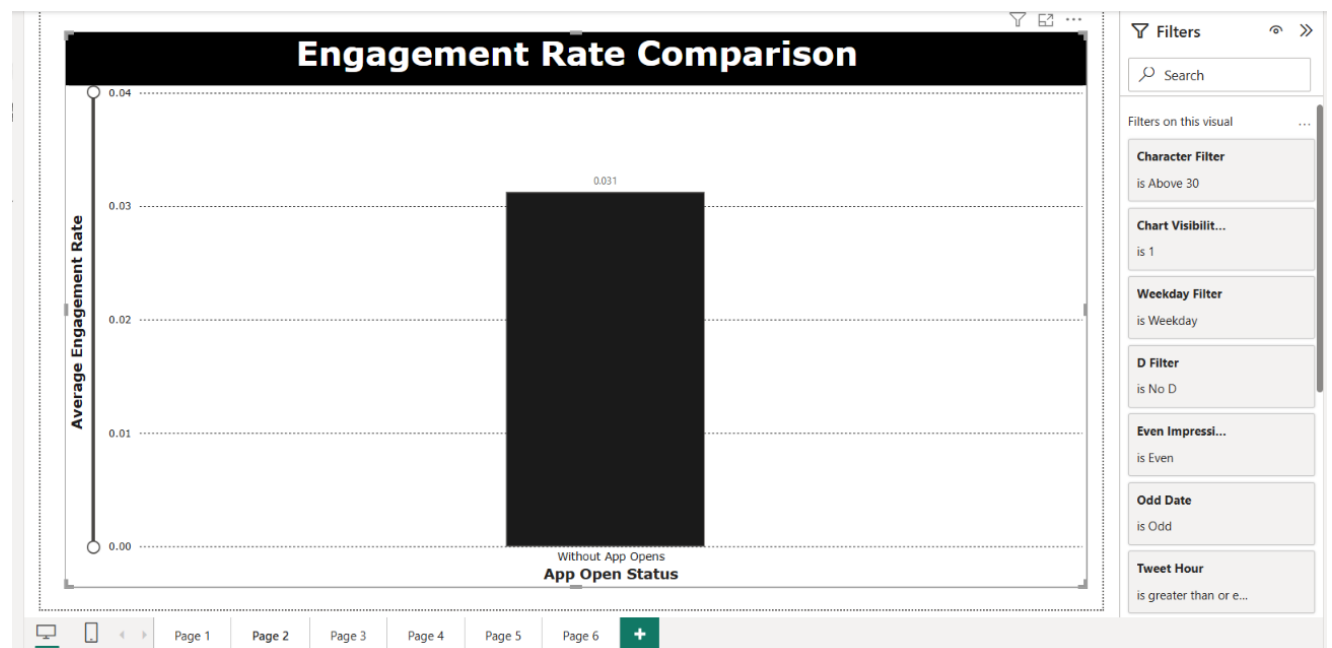
This task focused on comparing engagement rates for tweets based on application-opening activity available in the dataset.

The analysis included:

Engagement rate , Impressions , Tweet date , Character count , Weekday filtering.



Task (2) Without Filter



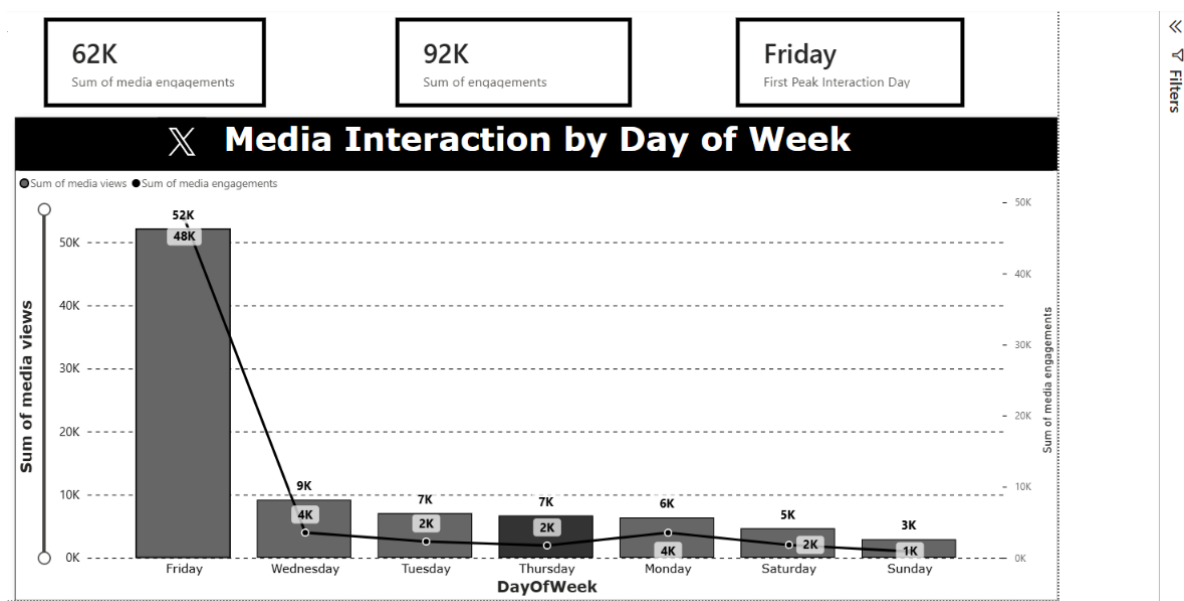
Task (2) With Filter

Task 3 – Media Interaction by Day of Week

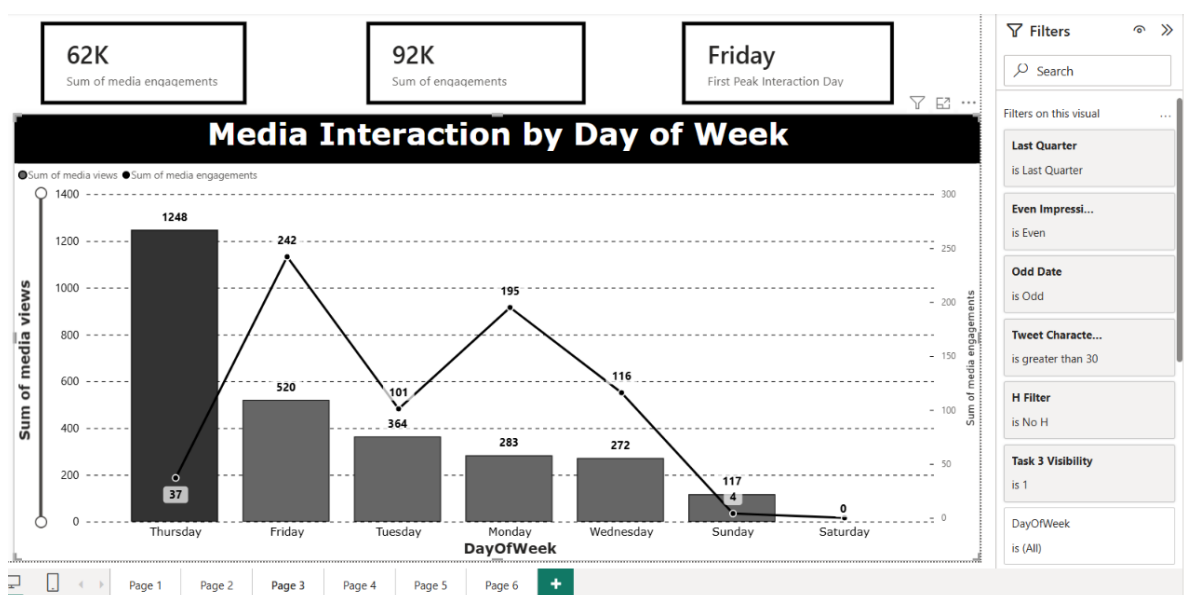
A visualization was created to compare **media views** and **media engagements** across different days of the week.

The analysis involved:

Media views , Media engagements , Day-of-week analysis , Last-quarter filtering , Impression and date conditions , Character-count filtering , Exclusion of tweets containing the specified character .



Task (3) Without Filter



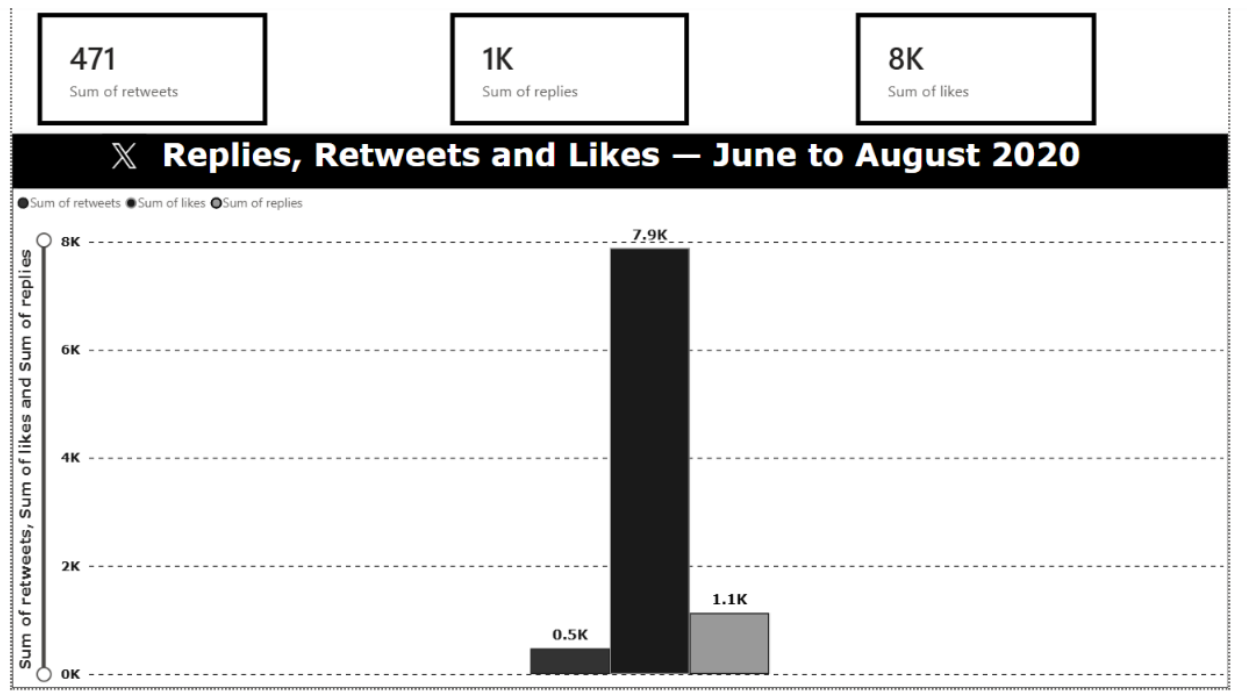
Task (3) With Filter

Task 4 – Replies, Retweets, and Likes Comparison

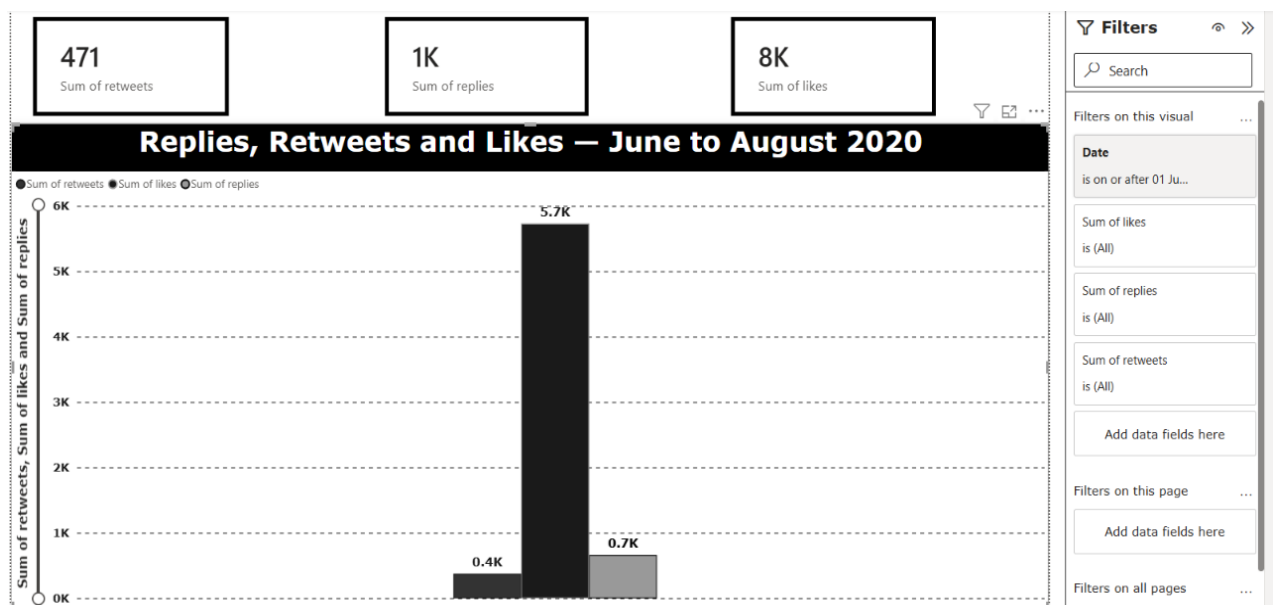
A bar chart was developed to compare overall:

Replies , Retweets , Likes.

The metrics were aggregated using **SUM** and filtered to include tweets posted between **June and August 2020**.



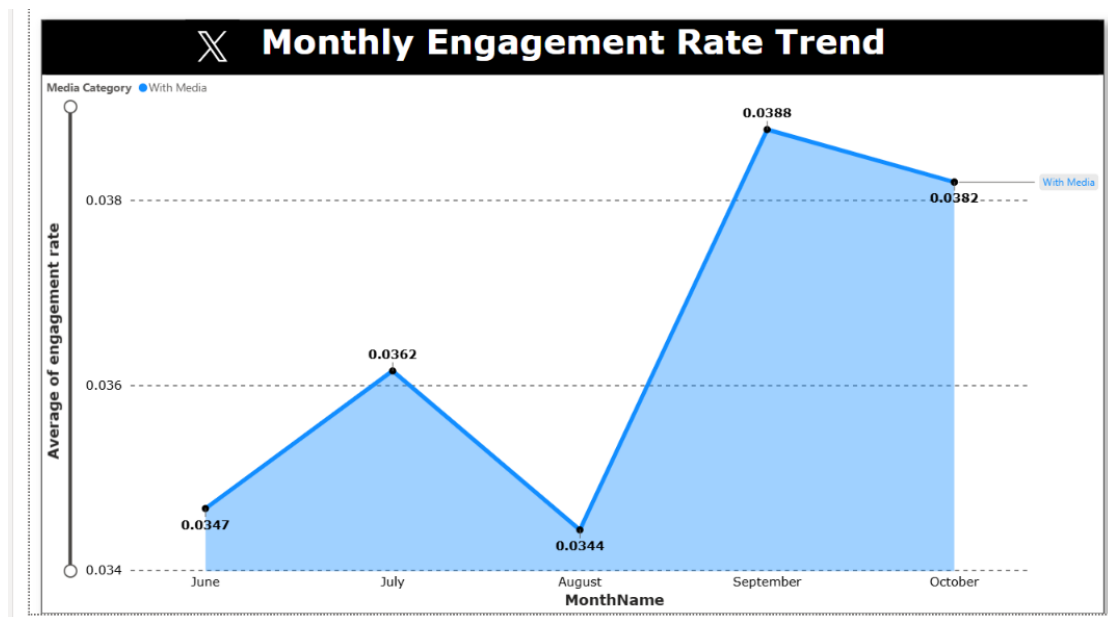
Task (4) Without Filter



Task (4) With Filter

Task 5 – Monthly Engagement Rate Trend

- A line chart was created to analyze the **average engagement rate by month**.
- The task focused on **identifying changes and trends in engagement over time**. The available media-related categories in the dataset were analyzed and used where applicable.



Task (5)

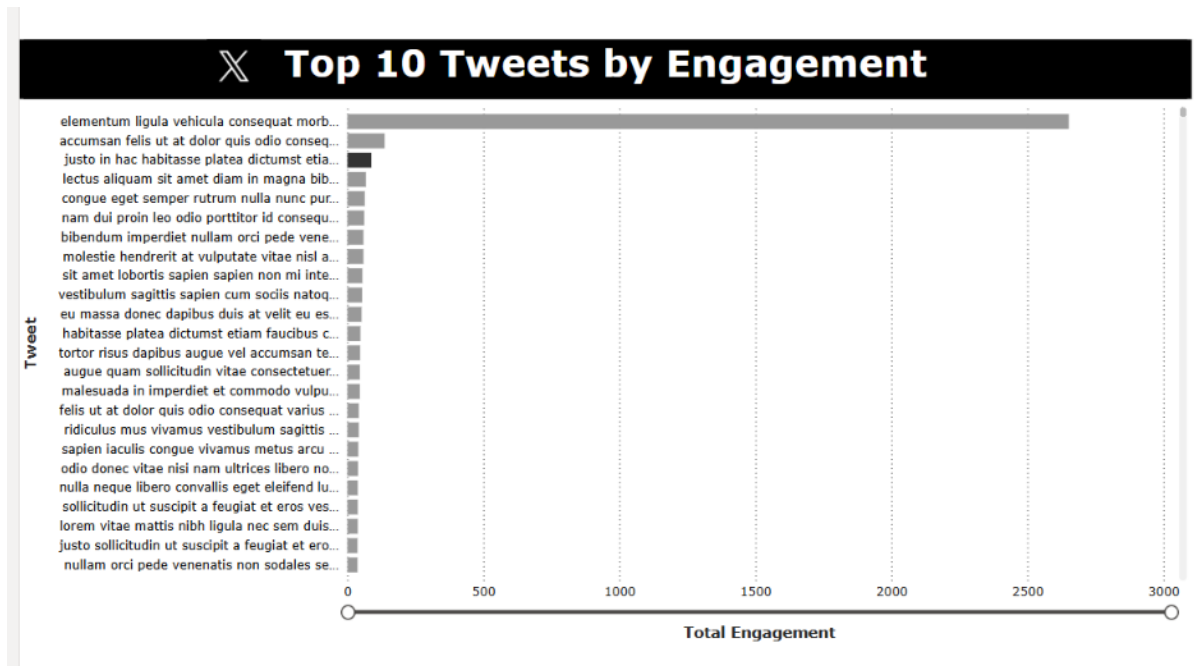
Task 6 – Top 10 Tweets by Engagement

The final task focused on identifying the **top 10 tweets based on the combined number of retweets and likes**.

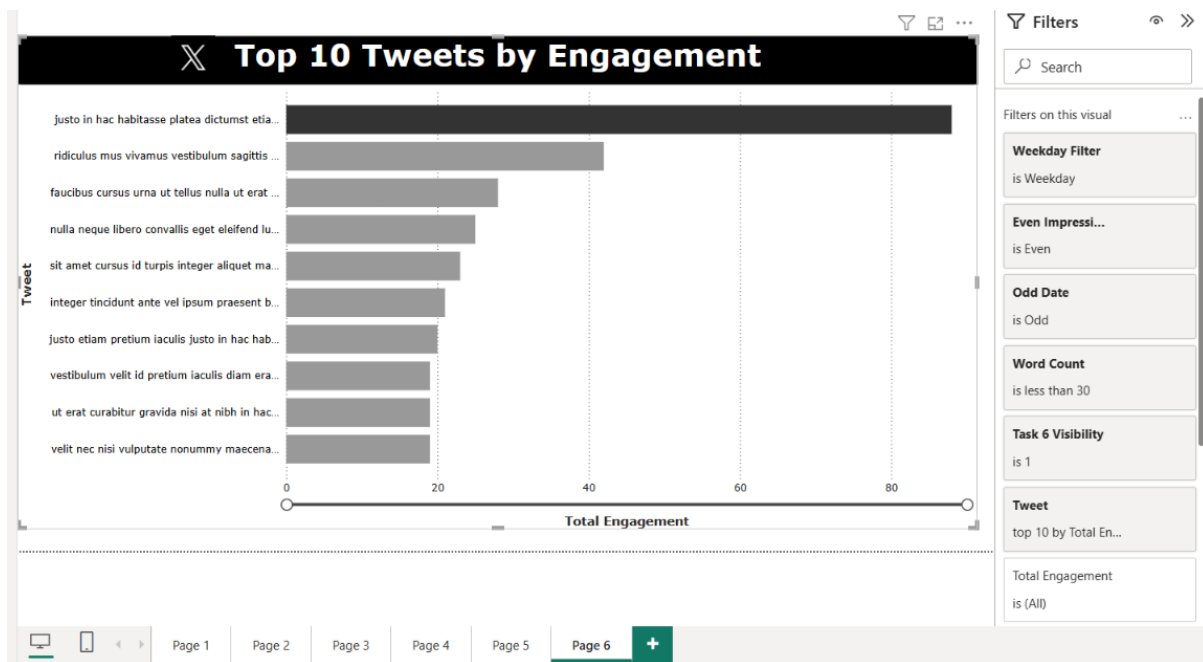
The analysis included:

Retweets , Likes , Combined engagement , Top 10 ranking , Weekday filtering , Impression filtering , Tweet-date filtering , Tweet word-count filtering .

Since a separate **user profile field was not available in the provided dataset**, the analysis was performed using the available tweet-level fields rather than inventing a profile field.



Task (6) Without Filter



Task (6) With Filter

Skills

Through this project, I developed and improved the following skills:

Technical Skills

- Microsoft Power BI
- Data import and preparation
- Data filtering
- Data aggregation
- DAX
- Date and time analysis
- Conditional measures
- Interactive dashboards

Analytical Skills

1. Identifying relevant data fields
2. Understanding social media engagement metrics
3. Comparing performance indicators

Feedback

The project provided practical experience in developing analytics reports using Power BI. Each task was implemented according to its specified analytical requirements, and the resulting visualizations were checked to ensure that the appropriate filters, aggregations, and conditions were applied.

Challenges and Solutions

During the project, several challenges were encountered while working with Power BI.

Challenge 1: DAX Errors

Some DAX formulas initially generated syntax and function errors.

Solution:

The formulas were reviewed and corrected by checking variable names, IF conditions, and DAX syntax.

Challenge 2: Filtering Data Correctly

Applying multiple conditions simultaneously, such as odd/even dates, impression conditions, character counts, and word counts, was challenging.

Solution:

Power BI's filter pane and DAX measures were used together to apply and validate the required conditions.

Challenge 3: Time-Based Visualization Visibility

Some visualizations were required to appear only during specific time periods.

Solution:

DAX-based visibility measures were created to determine whether the current time fell within the specified period.

Outcomes

The project successfully provided practical experience in **social media data analytics and Power BI dashboard development.**

The major outcomes were:

- ✓ Developed six Power BI analytical tasks.
- ✓ Created multiple interactive visualizations.

- ✓ Applied real-world filtering conditions.
- ✓ Developed DAX measures for calculations and conditional visibility.

Conclusion

- The Twitter Data Analytics project provided valuable hands-on experience in **Power BI, DAX, data analysis, and visualization**. By completing six different analytical tasks, I gained a better understanding of how raw social media data can be transformed into useful insights.
- The project helped me develop technical and analytical skills, particularly in **data filtering, aggregation, time-based analysis, dashboard creation**, and problem-solving. Working with different conditions and dataset limitations also improved my ability to analyze data carefully and make appropriate decisions based on the available information.
- Overall, this project was a valuable learning experience that strengthened my foundation in **Data Analytics and Business Intelligence** and provided practical exposure to tools and techniques that can be applied to real-world analytical problems.